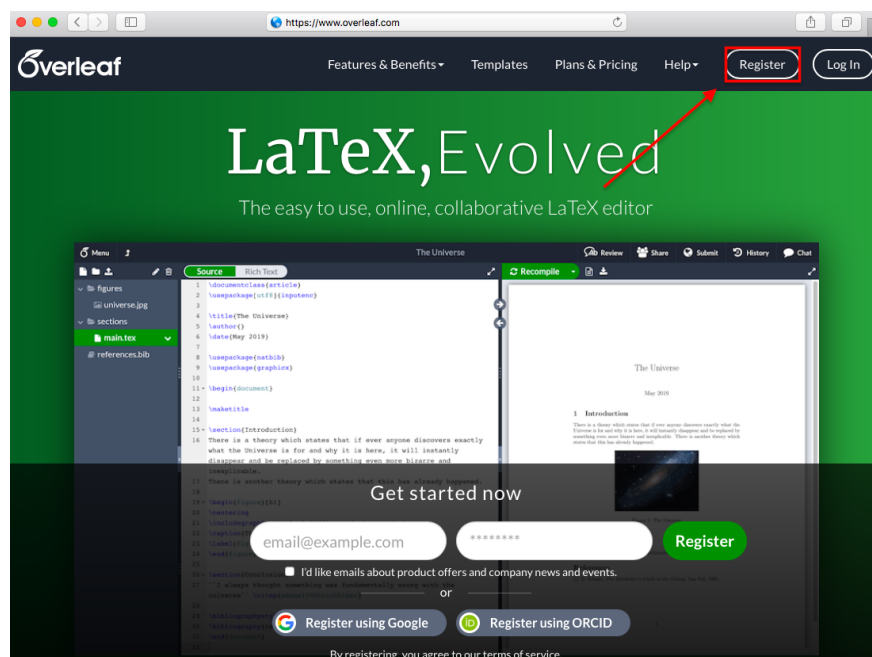


What is L^AT_EX?

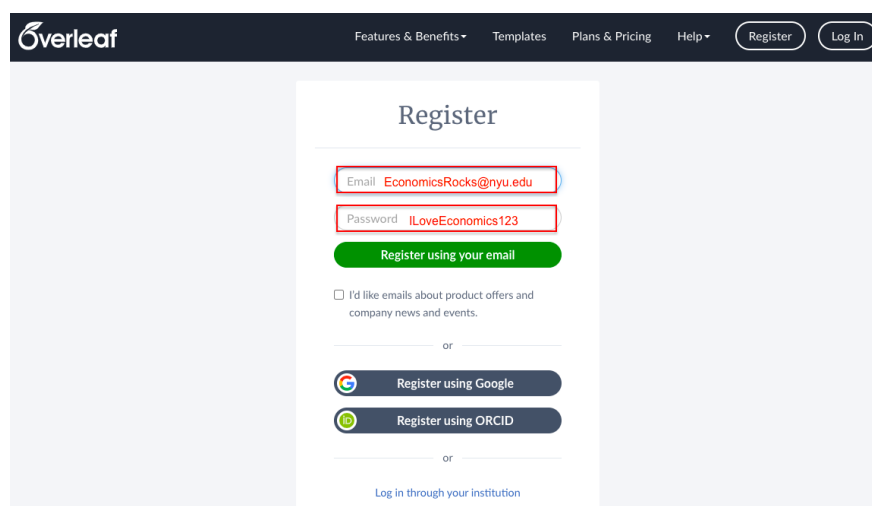
Latex¹ is a typesetting tool used to make high-quality, professional looking documents. It was created by Leslie Lamport in the early 1980s and has many advantages (depending on the application) over other typesetting options, like Microsoft Word. Its easy to change formatting and interface that allows the user to write equations effortlessly are some reasons it is so commonly used in science, mathematics, and economics.

Getting Started

While there are many L^AT_EX editors, I recommended you use Overleaf for this project. Start by going to <https://www.overleaf.com> and click register (see figure below).

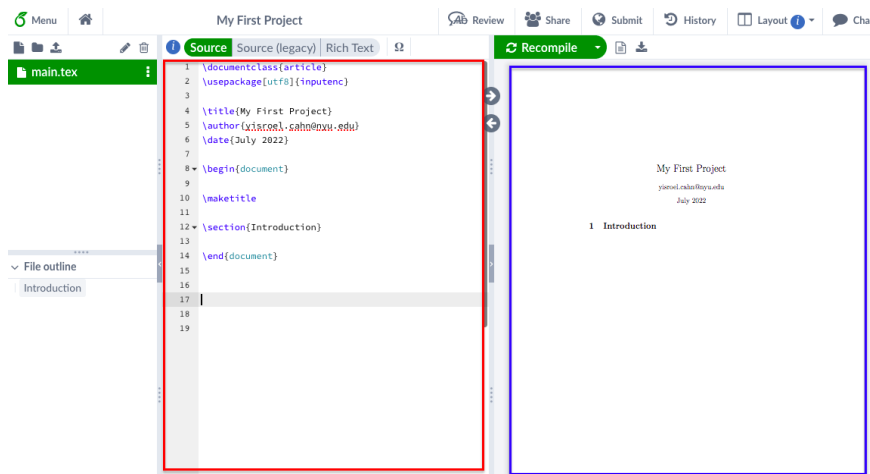


Use an email address and make up a password (see figure below).

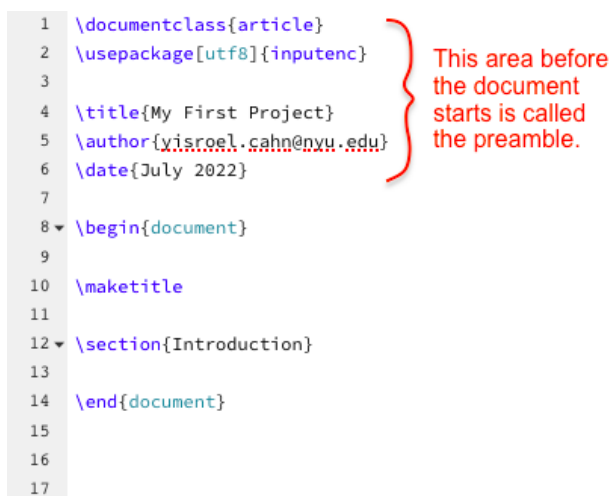


¹Latex is stylized as L^AT_EX and pronounced “Lay-tek” because the characters ‘T’, ‘E’, and ‘X’ in Latex are meant to be the Greek letters tau, epsilon, and chi.

Click “Create First Project” and choose “Blank Project.”² When prompted, name the project “My First Project”. Your screen should look like this:



Source code (what you write to make your document) is typed into the red box, it is compiled (turned into language the computer understands so it can make your document), and produces the PDF on the right in the blue box. Let’s talk about what each line of code is doing — take a look at the figures below



²Overleaf has many different templates for presentations, resumes, and more, but for now we are just writing a simple article. Overleaf also allows you to collaborate with others, much like a Google doc. It even allows you to revert the document to an old version.

```
1 \documentclass{article}
2 \usepackage[utf8]{inputenc}
3
4 \title{My First Project}
5 \author{Yisroel Cahn@nyu.edu}
6 \date{July 2022}
7
8 \begin{document}
9
10 \maketitle
11
12 \section{Introduction}
13
14 \end{document}
15
16
17
```

This line in the preamble says what kind of document it is. In this case, the document is an article.

```
1 \documentclass{article}
2 \usepackage[utf8]{inputenc}
3
4 \title{My First Project}
5 \author{Yisroel Cahn@nyu.edu}
6 \date{July 2022}
7
8 \begin{document}
9
10 \maketitle
11
12 \section{Introduction}
13
14 \end{document}
15
16
17
```

This line adds a package which enables the document to have additional features. We will add more packages later.

```
1 \documentclass{article}
2 \usepackage[utf8]{inputenc}
3
4 \title{My First Project}
5 \author{Yisroel Cahn@nyu.edu}
6 \date{July 2022}
7
8 \begin{document}
9
10 \maketitle
11
12 \section{Introduction}
13
14 \end{document}
15
16
17
```

These lines define features of the document's title.

```
1 \documentclass{article}
2 \usepackage[utf8]{inputenc}
3
4 \title{My First Project}
5 \author{yisroel.cahn@nyu.edu}
6 \date{July 2022}
7
8 \begin{document}
9
10 \maketitle
11
12 \section{Introduction}
13
14 \end{document}
```

← This line starts the document.

← This line ends the document.

```
1 \documentclass{article}
2 \usepackage[utf8]{inputenc}
3
4 \title{My First Project}
5 \author{yisroel.cahn@nyu.edu}
6 \date{July 2022}
7
8 \begin{document}
9
10 \maketitle
11
12 \section{Introduction}
13
14 \end{document}
```

← Now that we are finally inside the document, we start by printing the title (we already defined its features in the preamble).

```
1 \documentclass{article}
2 \usepackage[utf8]{inputenc}
3
4 \title{My First Project}
5 \author{yisroel.cahn@nyu.edu}
6 \date{July 2022}
7
8 \begin{document}
9
10 \maketitle
11
12 \section{Introduction}
13
14 \end{document}
```

← This is a section in the document, which we are calling "Introduction."

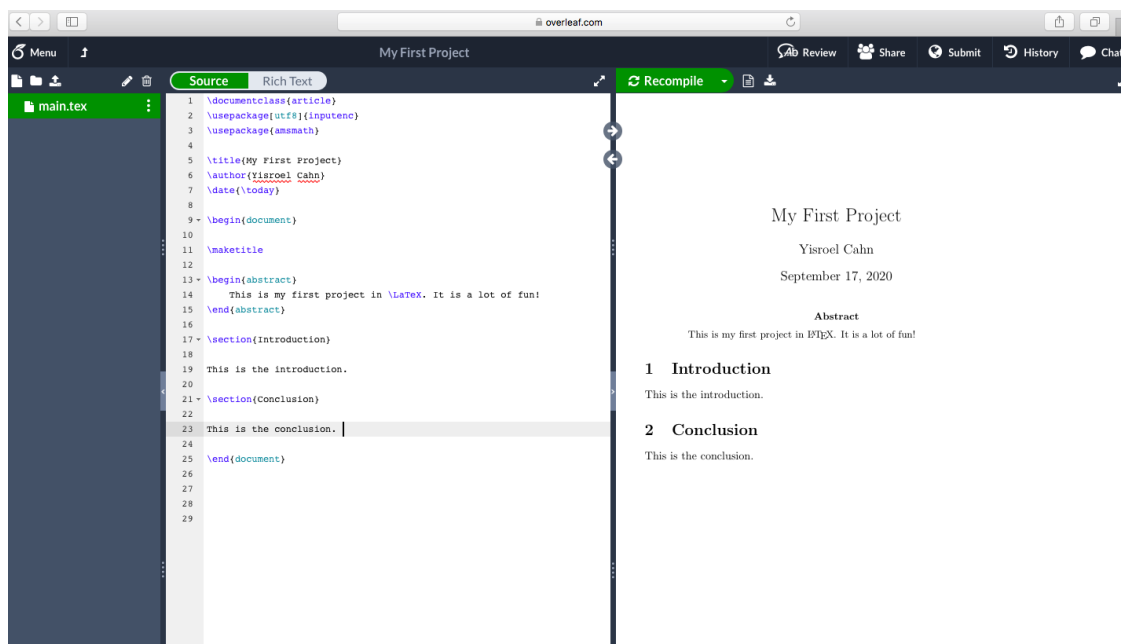
Let's start by doing a couple of things: change your name from you email address to your actual name, change the date to `\today`, add `\usepackage{amsmath}` in the preamble under `\usepackage[utf8]{inputenc}`, add

```
\begin{abstract}
```

This is my first project in \LaTeX. It is a lot of fun!

```
\end{abstract}
```

under `\maketitle`, type “This is the introduction.” in the introduction section, and add a section called “Conclusion” (by writing `\section{Conclusion}` after the introcuton section) with the word “This is the conclusion.” in it. After you press `ctrl+s` or hit “Recompile,” your screen should look like this



Basic Commands and Math Mode

Commands in Latex are done using the backslash button (`\`). Create a section in between the introduction and the conclusion called “Basic Commands and Math Mode.” Then type

The word `\textit{hello}` is italicized. And now, the word `\textbf{hello}` is in bold.

Notice that things inside the `{}` are the arguments of the command. A common command is to begin and end an object. For example, type

Bullet points look like

```
\begin{itemize}
```

```
\item bullet point 1
```

```
\item bullet point 2
```

```
\item bullet point 3
```

```
\end{itemize}
```

and lists look like

```
\begin{enumerate}
```

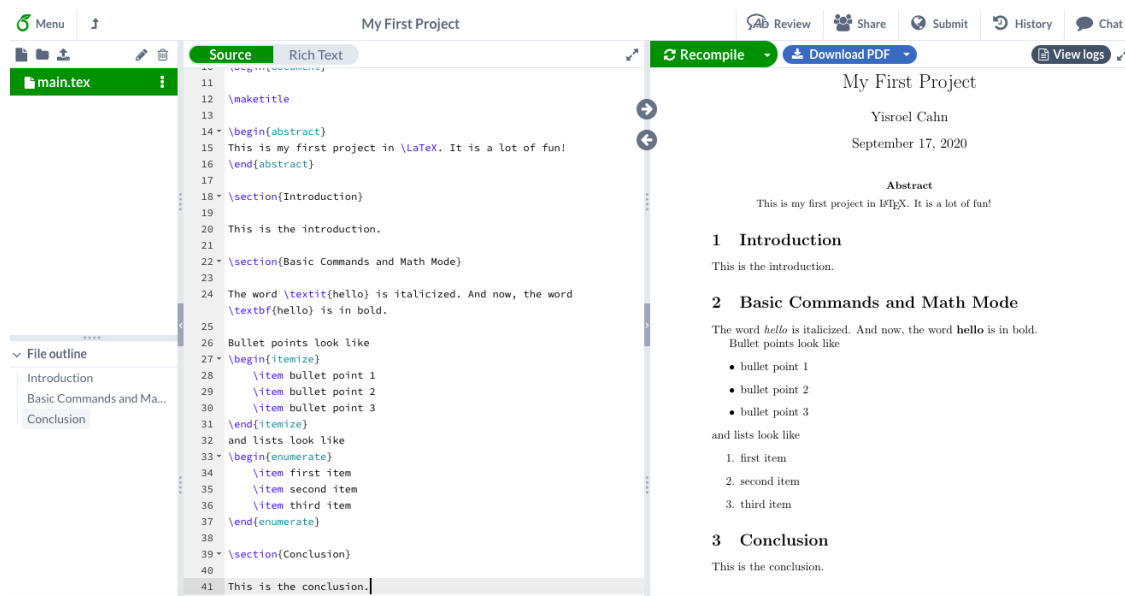
```
\item first item
```

```
\item second item
```

```
\item third item
```

```
\end{enumerate}
```

in the same section. Your screen should look like this

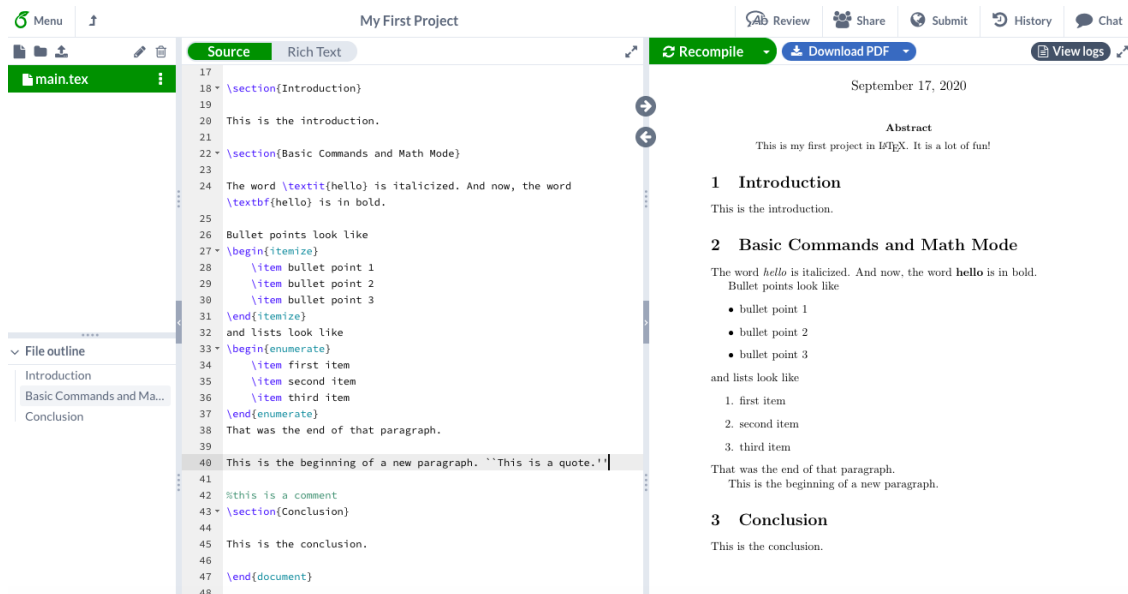


If you want to write something in the source code to explain what you are doing (for someone else who might look at your source code or for yourself for future reference so you don't forget), you can write a comment. Comment by using the `%`. For example write `%this is a comment` above `\section{Conclusion}`. New paragraphs are started by leaving a space between paragraphs in the source code. Continuing from where we left off, add

```
...
\item third item
\end{enumerate}
That was the end of that paragraph.
```

This is the beginning of a new paragraph.

Quotations are done using two back-ticks and two apostrophes (`''''`) and single quotations are done with one back-tick and apostrophe (`''`). Add `''This is a quote.''` to the end of the last paragraph. Your screen should look like



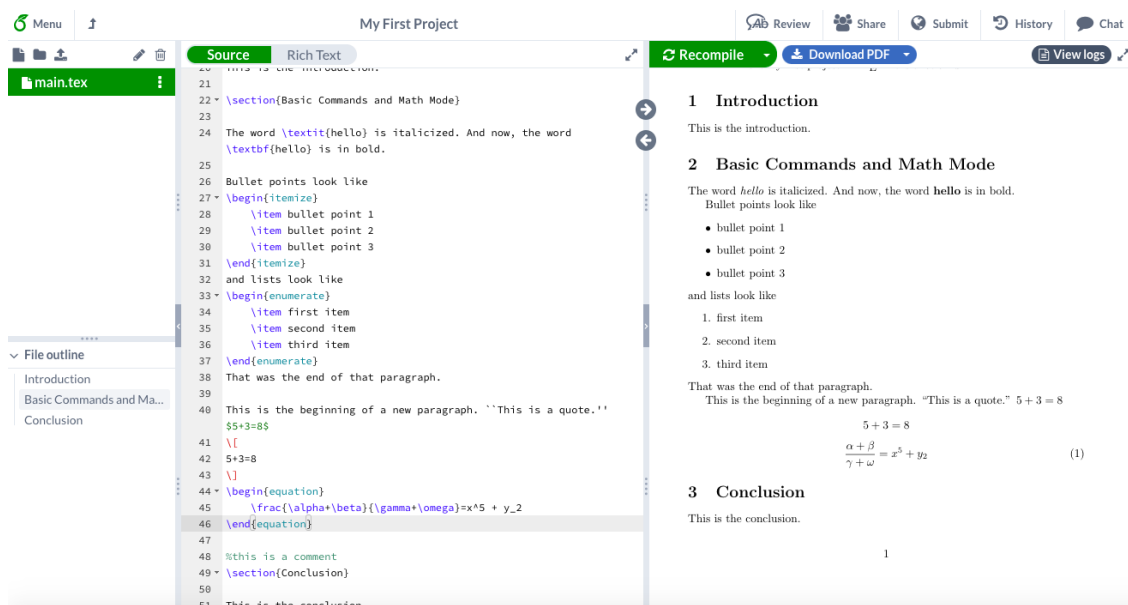
Latex has a separate mode for entering math equations. To enter math mode in-line use `$` and exit in-line math mode also with `$`. To enter and exit math mode on a new line use `\[ \]`. Add `$5+3=8$` to the end of the last paragraph of Section 2 and

```
\[
5+3=8
\]
```

right after. To write an equation, some math symbols, superscripts, subscripts, and fractions, type for example

```
\begin{equation}
\frac{\alpha+\beta}{\gamma+\omega}=\gamma^5 + y_2
\end{equation}
```

Your screen should look like



`\frac` is how to write fractions in math mode with the first argument in `{}` being the numerator and the second argument in the second `{}` being the denominator. Try typing

$$\frac{1}{2} + \left(\frac{\alpha}{2}\right)^2 = \frac{2 + \alpha^2}{4}$$

in the same section and note that in order to make the parenthesis larger, you need to write “`\left(`” and “`\right)`” so that the parentheses becomes as large as what is in-between them.

Next, in the same section, try typing

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

by googling how to do the plus-minus sign and square root sign in Latex.

To produce aligned equations like

$$\begin{aligned} y &= 6 + 7 \\ &= (3 + 3) + (4 + 3) \\ &= (3 + 2 + 1) + (2 + 2 + 3) \end{aligned}$$

type

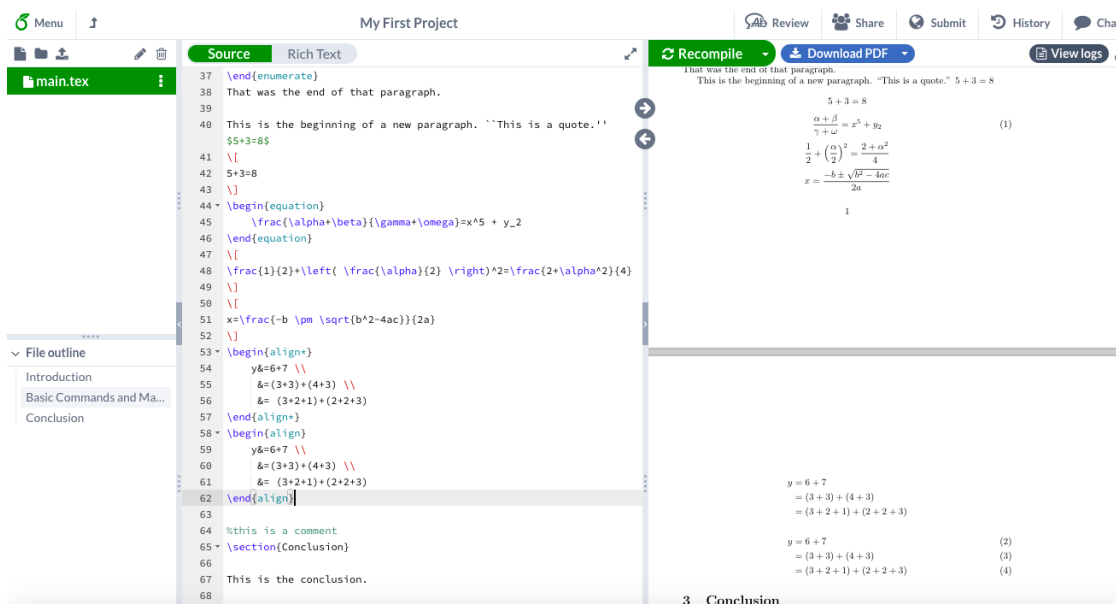
```
\begin{align*}
y&=6+7 \\
&=(3+3)+(4+3) \\
&= (3+2+1)+(2+2+3)
\end{align*}
```

in the same section, and note that `\\` starts a new line and `&` is where the equation aligns. Add

```
\begin{align}
y&=6+7 \\
&=(3+3)+(4+3) \\
&= (3+2+1)+(2+2+3)
\end{align}
```

in the same section and note that it puts equations numbers after each line.

Your screen should look like



Referencing

Make a new section before the conclusion called referencing. In it, create two subsections by typing

```
\subsection{Cross-Referencing}
```

```
\subsection{Bibliography}
```

Next to reference the section “Referencing”, type `\label{sec:Referencing}` after the section, like

```
\section{Referencing} \label{sec:Referencing}
```

Under the subsection “Cross-Referencing” type

You can reference a section if it is labeled by typing `\ref{sec:Referencing}`

(whatever is inside the label) and you can reference an equation like this

```
\begin{equation} \label{eq:Euler}
```

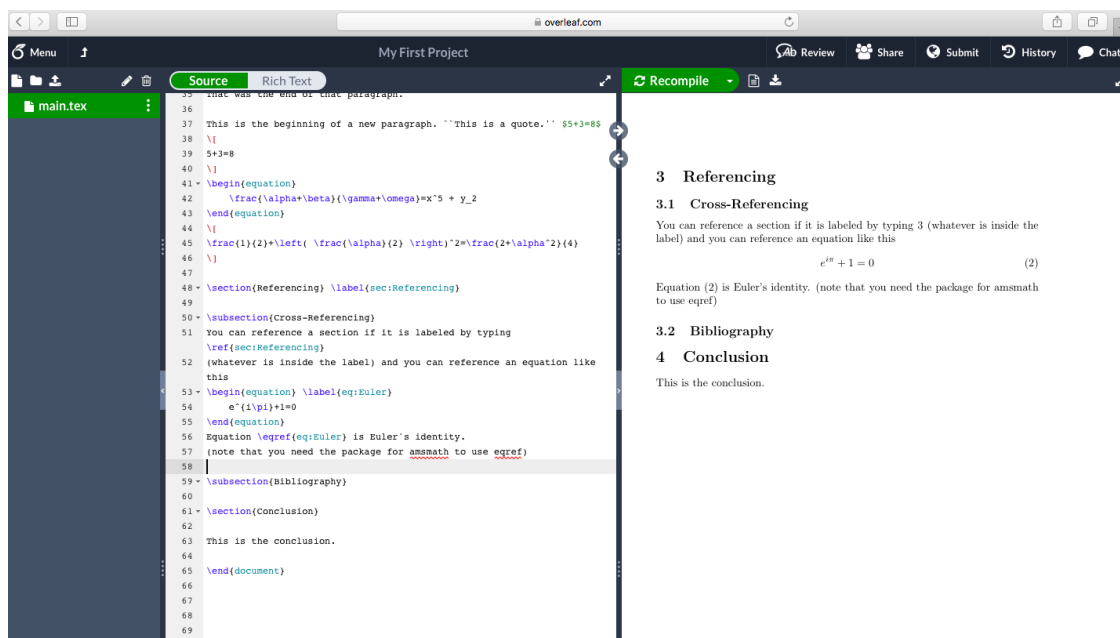
$$e^{i\pi} + 1 = 0$$

```
\end{equation}
```

Equation `\eqref{eq:Euler}` is Euler’s identity.

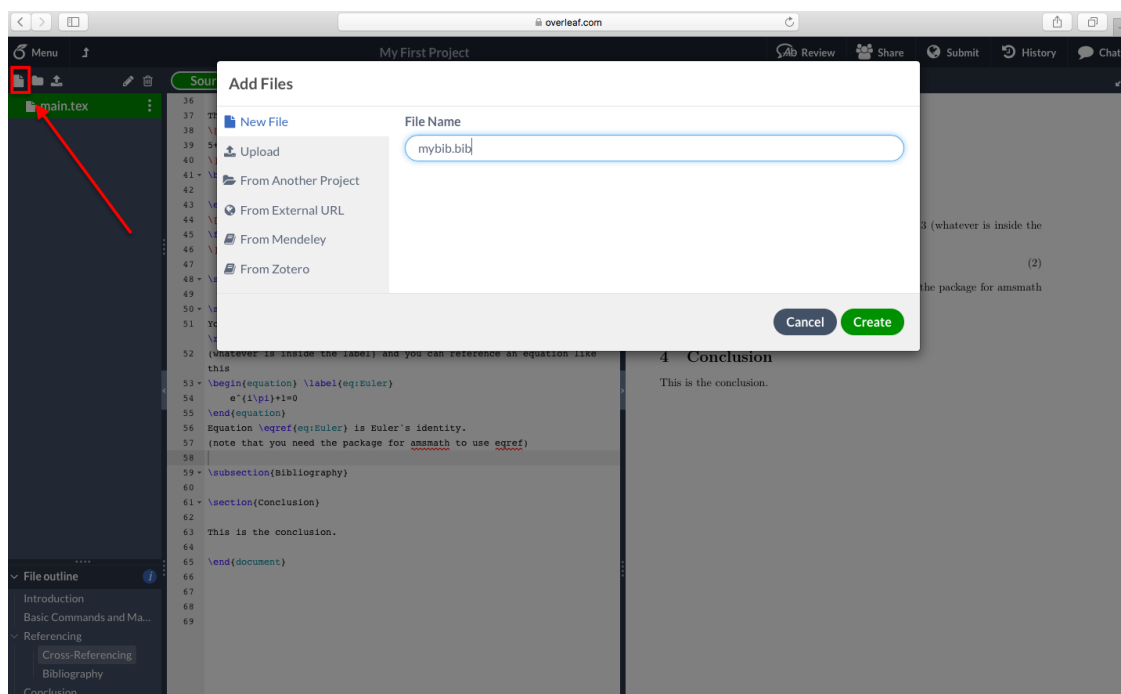
(note that you need the package for `amsmath` to use `eqref`)

Your screen should look like



Cross-referencing in this way is extremely useful. While writing a paper, you will be moving around sections, equations, figures, and tables (which we will talk about shortly) and having to manually change each cross-reference would be a nightmare!

Now we are going to add a bibliography to our document. Click the page button and create a new file called “mybib.bib” (as shown in the figure below).



In the new file, type

```
@article{atkinson1982,
  author = {Atkinson, Anthony B. and Bourguignon, Francois},
  title  = {The Comparison of Multi-Dimensioned Distributions of Economic Status},
  journal = {Review of Economic Studies},
  volume = {49},
  number = {2},
  pages  = {183--201},
  year   = {1982}
}
```

Save the file (Ctrl+S on Windows/Linux or Cmd+S on Mac) and return to `main.tex`.

In the preamble, add

```
\usepackage{natbib}
\bibliographystyle{apalike}
```

The first line loads the `natbib` package, which provides flexible citation commands and is the standard bibliography package used by most economics journals. The second line specifies the bibliography style. Many journals provide their own bibliography style file, but `apalike` is a good default for this tutorial. At the end of your document, immediately before `\end{document}`, add

```
\bibliography{mybib}
```

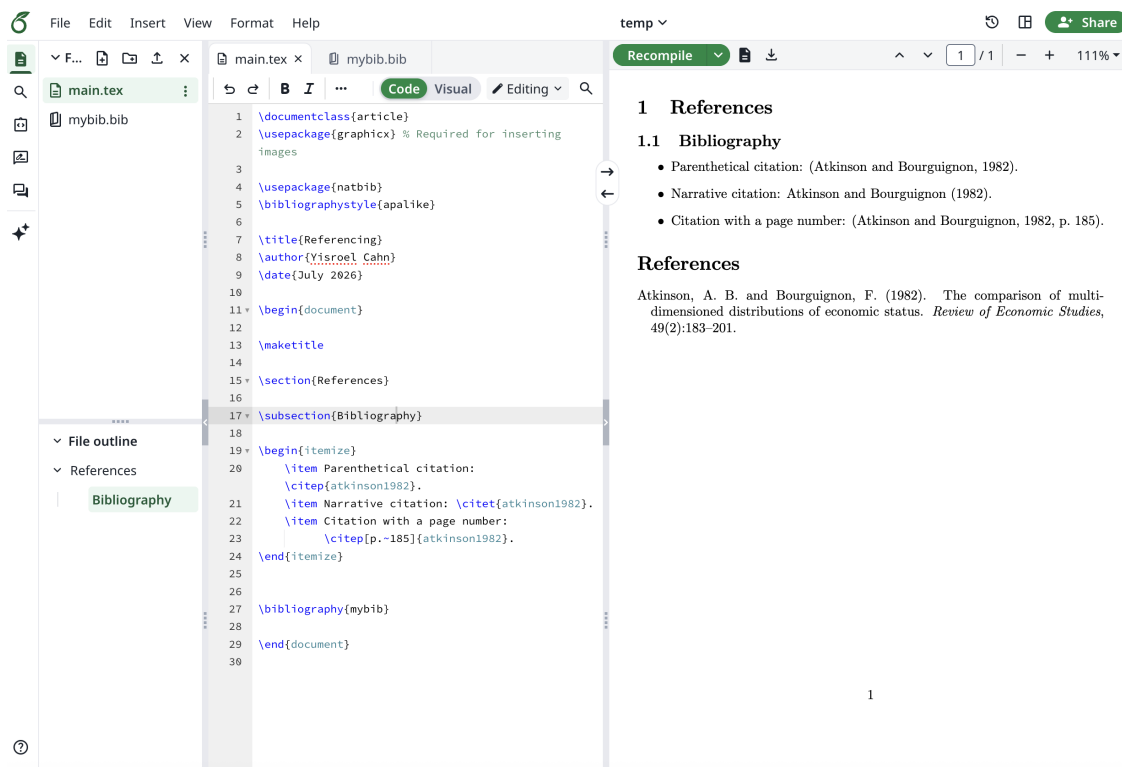
Now, under the subsection “Bibliography”, type

```
\begin{itemize}
  \item Parenthetical citation: \citep{atkinson1982}.
  \item Narrative citation: \citet{atkinson1982}.
  \item Citation with a page number:
    \citep[p.~185]{atkinson1982}.
\end{itemize}
```

3

Your project should now look similar to the figure below.

³I recommend keeping your `.bib` file in alphabetical order and using descriptive citation keys such as “authoryear” (e.g., `atkinson1982` or `maasoumi1986`).



To make citations and references clickable, add

`\usepackage{hyperref}`

near the end of your preamble (after most other packages).

Unlike `biblatex`, `natbib` separates citation commands from bibliography formatting. When submitting a paper to an economics journal, you will typically replace

`\bibliographystyle{apalike}`

with the bibliography style required by the journal.

Now we are going to import a citation directly from the University of Pennsylvania Library. Go to

<https://www.library.upenn.edu/>

or search Google for “UPenn Library EconLit.” Sign in with your PennKey if prompted, open **EconLit**, and search for

“The Measurement and Decomposition of Multi-dimensional Inequality.”

Your screen should look similar to the figure below.

Your search found results in **Articles+** [View all 2,503 Articles+ results](#)

1. [The Measurement and Decomposition of Multi-dimensional Inequality](#)
Esfandiar Maasoumi, *The Economic Theory of Income Inequality*. 2013, pp. 447-53, 2013
[Book Chapter](#)
2. [THE MEASUREMENT AND DECOMPOSITION OF MULTI-DIMENSIONAL INEQUALITY: 1. INTRODUCTION](#)
Esfandiar Maasoumi, *Econometrica*, 1986
[Journal Article: Full text online](#)
3. [The Measurement and Decomposition of Multi-Dimensional Inequality](#)
Esfandiar Maasoumi, *Econometrica*, 1986
[Journal Article: Full text online](#)

[Journal Article \(1,757\)](#) [Dissertation \(519\)](#) [Book Chapter \(113\)](#) [Conference Proceeding \(61\)](#)
[Book / eBook \(19\)](#) [Paper \(14\)](#)

Open the article's record (or the full text if available), click the **Cite** button, and select **BibTeX** as the export format. Your screen should look similar to the figure below.

Cite



Copy citations directly into your paper

BibTeX

```
@article{
  author={Maasoumi,Esfandiar},
  year={1986},
  month={07},
  title={THE MEASUREMENT AND DECOMPOSITION OF MULTI-DIMENSIONAL
  INEQUALITY: 1. INTRODUCTION},
  journal={Econometrica (1986-1998)},
  volume={54},
  number={4},
  pages={991},
  note={Copyright - Copyright Econometric Society Jul 1986; Last updated - 2024-11-
  10; CODEN = ECMTA3}
```

RefWorks

Copy

Consult RefWorks [style guidelines](#) to check the accuracy and completeness of your citations.

Export to a citation manager or file



RefWorks



EndNote

Zotero, Citavi
etc.

NoodleTools



EasyBib

Microsoft
Excel Format

Done

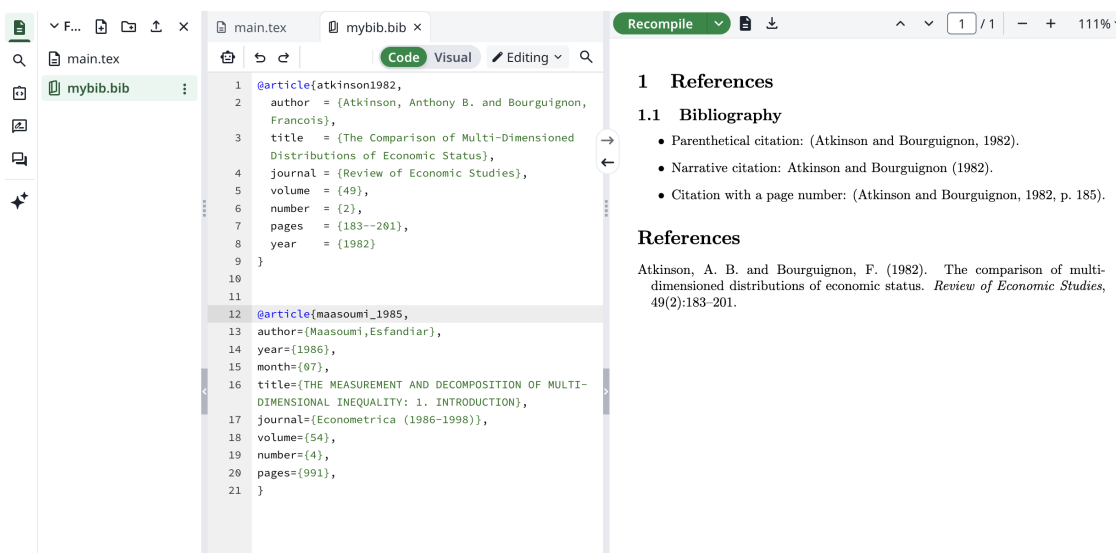
Copy and paste the BibTeX entry into your `mybib.bib` file. It is often helpful to simplify the citation key. For example, you might change

```
@article{LONG_DATABASE_NAME,...}
```

to

```
@article{maasoumi1986,...}
```

You can also remove unnecessary fields such as `url`, `abstract`, and database-specific identifiers. Your cleaned BibTeX entry should resemble the figure below.



Notice that simply adding an entry to your `mybib.bib` file does not make it appear in the bibliography. A reference will only appear if it is cited somewhere in your document.

If you would like to include a reference without citing it in the text, add

```
\nocite{maasoumi1986}
```

immediately before

```
\bibliography{mybib}
```

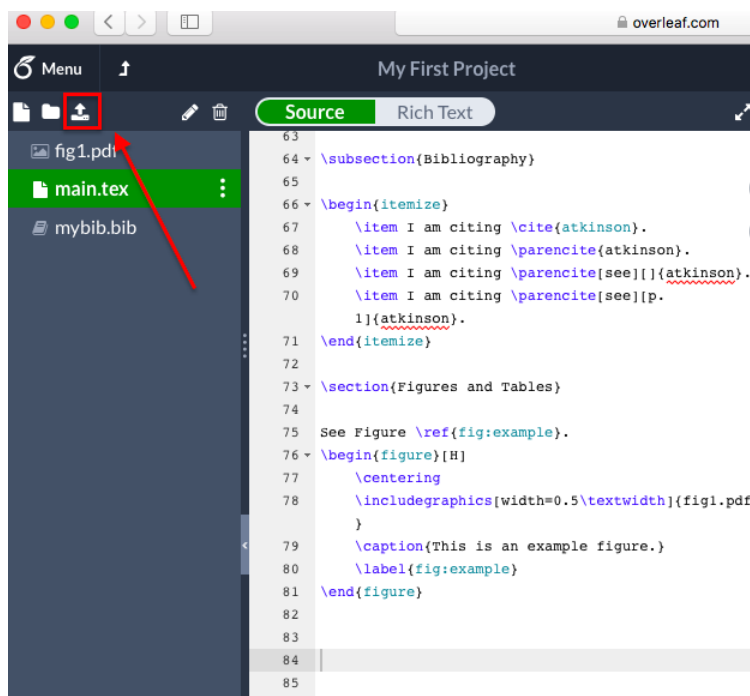
To include every entry in your bibliography file, use

```
\nocite{*}
```

This is useful for checking that all of your BibTeX entries are formatted correctly before submitting a paper.

Figures and Tables

After downloading `example.png` from Brightspace, upload it to your project by click the upload button in Overleaf. See the figure below



Add `\usepackage{graphicx}` and `\usepackage{float}` to the preamble. Add a section called “Tables and Figures” before the conclusion section. In the new section type

See Figure `\ref{fig:example}`.

`\begin{figure}[H]`

`\centering`

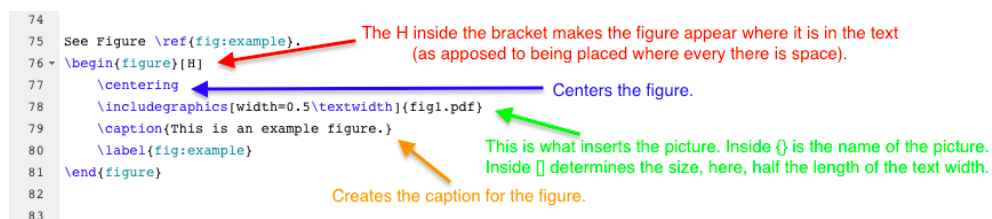
`\includegraphics[width=0.5\textwidth]{example.png}`

`\caption{This is an example figure.}`

`\label{fig:example}`

`\end{figure}`

Let’s break down what is going on. See the figure below



Next let’s make a table. In the same section type

This is a table

`\begin{table}[H]`

`\centering`

`\begin{tabular}{l l}`

`1 & 2 \\\`

`3 & 4`

`\end{tabular}`

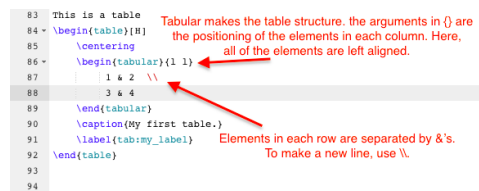
`\caption{My first table.}`


```

\label{tab:my_label}
\end{table}

```

The output should be



```

1 2
3 4

```

Table 1: My first table.

5 Conclusion

This is the conclusion.

To get the gist of it, add the following two tables

This is another table

```

\begin{table}[H]
\centering
\begin{tabular}{l c r} \hline \hline
left & center & right \\\ \hline
1 & 22 & 333 \\  

4444 & 55555 & 66666666 \\  

777 & 88 & 9999 \\\ \hline
\end{tabular}
\caption{My second table.}
\label{tab:two}
\end{table}

```

and

This is another table

```

\begin{table}[H]
\centering
\begin{tabular}{|l|l|l|l|} \hline
$\alpha$ & $\beta$ & $\gamma$ \\\ \hline
123 & 353 & 94837 \\  

45734 & 54557 & 66 \\  

67783 & 37 & 1899 \\\ \hline
\end{tabular}
\caption{My third table.}
\label{tab:three}
\end{table}

```

Last, add this table

Name	Age	Height	Weight
Adam	20	5'9"	165 lb
Benjamin	24	5'5"	152 lb
Carla	19	5'3"	127 lb
Deborah	27	5'6"	141 lb

Tikz

Tikz is a package in latex for drawing figures. Add `\usepackage{tikz}` to the preamble. A good website with lots of economics figures to tweak is <https://sites.google.com/site/kochiuyu/Tikz>.

Make a new section above the conclusion section called “Tikz”. In the new section, type

Figure \ref{fig:tikz} is a figure using the Tikz package.

```
\begin{figure}[H]
  \centering
  \begin{tikzpicture}[scale=0.3]
    \draw (5,0) -- (-5, 0);
  \end{tikzpicture}
  \caption{My first Tikz figure}
  \label{fig:tikz}
\end{figure}
```

This draws a line from coordinates (5,0) to (-5,0) and the picture’s size is “0.3”.

Now type

Figure \ref{fig:tikz2} is a figure using the Tikz package.

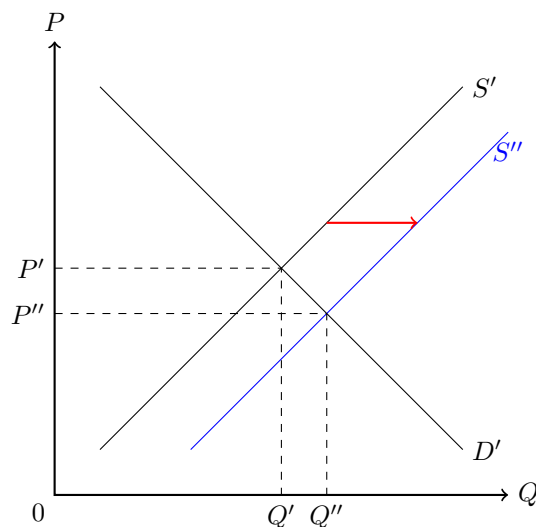
```
\begin{figure}[H]
  \centering
  \begin{tikzpicture}[scale=0.3]
    \draw (5,0) -- (-5, 0);
    \draw[>-] (0,0) -- (5,5);
    \draw[red, dashed] (-1,0) -- (-1,7);
    \node [left] at (-2,3.5) {$W_1$};
    \filldraw [blue] (0,0) circle (4pt);
  \end{tikzpicture}
  \caption{My second Tikz figure}
  \label{fig:tikz2}
\end{figure}
```

and

Figure \ref{fig:tikz3} is a figure using the Tikz package.

```
\begin{figure}[H]
  \centering
  \begin{tikzpicture}[scale=0.6]
    \draw[thick,<->] (0,10) node[above]{$P$}--(0,0)--(10,0) node[right]{$Q$};
    \node [below left] at (0,0) {$0$};
    \node [below] at (5,0) {$Q^*$};
    \node [left] at (0,5) {$P^*$};
    \draw(1,1)--(9,9) node[right]{\texttt{Supply}};
    \draw(1,9)--(9,1) node[right]{\texttt{Demand}};
    \draw[dashed] (0,5)--(5,5)--(5,0);
  \end{tikzpicture}
  \caption{My third Tikz figure.}
  \label{fig:tikz3}
\end{figure}
```

If you would like (optional), add this Tikz figure to the section



note that $P^{\prime}$ produces P'' .

Beamer

To make slides using latex, the document type is called beamer. This is beyond the scope of this introduction, but interested readers should look at [https://www.overleaf.com/learn/latex/Free_online_introduction_to_LaTeX_\(part_3\)](https://www.overleaf.com/learn/latex/Free_online_introduction_to_LaTeX_(part_3)).

Note on Errors

Don't panic! Errors happen. It is usually a misspelling. Recompile often and fix errors as soon as they pop up. It is much easier to fix one error than to try to fix several.

Google

Now that you know the basics, if there is anything new you want to do just Google it! Getting good at Latex takes a little while and you will be Googling a lot of things in the beginning, but as you get better, using Latex will be really efficient.

You are done!

You just made your first document in L^AT_EX! Click "Download PDF" (as shown in the figure below), and submit in on Brightspace.

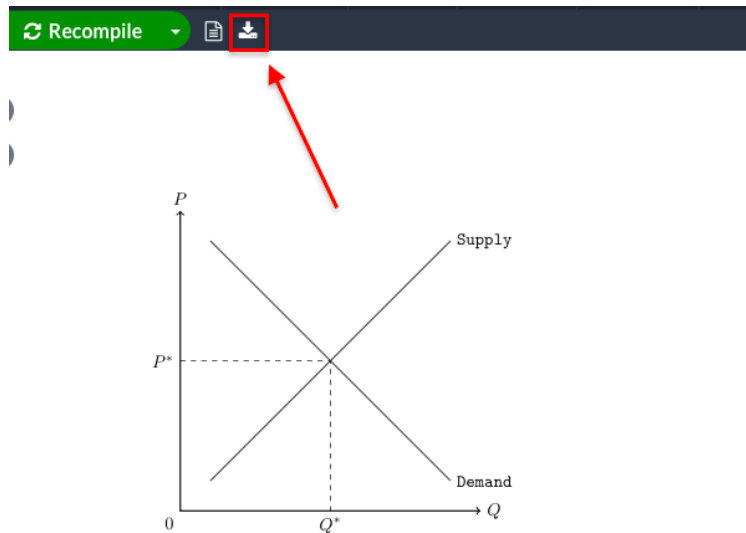


Figure 4: My third Tikz figure.

6 Conclusion

This is the conclusion.

References

- Atkinson, A. B., & Bourguignon, F. (1982). The comparison of multi-dimensioned distributions of economic status. *Review of Economic Studies*, 49(2), 183–201.
- Maasoumi, E. (1986). The measurement and decomposition of multi-dimensional inequality. *Econometrica*, 54(4), 991–997.

Word of Advice

Don't go crazy with L^AT_EX. It is useful in some settings and not the best choice in others. Undergrads tend to get carried away and write everything, even their shopping lists, with Latex. Unless it is an academic project that uses a sufficient amount of mathematical notation, L^AT_EX is likely not the best choice and something like Word might be preferred.